

Team Notebook

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1 Data Structure	2
1.1 2d fenwick tree	2
1.2 centroid decomposition	2
1.3 dynamic segment tree	2
1.4 fenwick tree	2
1.5 hld	2
1.6 hld edge	3
1.7 merge sort tree	3
1.8 multi ordered set	4
1.9 ordered set	4
1.10 persistent segment tree	4
1.11 persistent segment tree lazy	4
1.12 segment tree	5
1.13 segment tree 2d	5
1.14 segment tree beats	5
1.15 segment tree lazy	6
1.16 sparse table	6
1.17 trie bitmask	6
1.18 trie string	7
2 Geometry	7
2.1 convex hull trick	7
2.2 li chao tree	7
3 Graphs	7
3.1 2sat	7
3.2 bellman ford	8
3.3 blossom	8
3.4 dinic	8
3.5 dominator tree	9
3.6 floyd warshall	9
3.7 kuhn	9
3.8 lca	10

3.9 maxflow mincost	10
3.10 tree bridge	11
4 Math	11
4.1 cht	11
4.2 criba	11
4.3 diophantine	12
4.4 fft	12
4.5 fwht bitwiseconvolution	13
4.6 gauss	13
4.7 geo2d	13
4.8 matrix	14
4.9 millerrabin polardrho	14
4.10 ntt	15
4.11 primitive root	15
4.12 random	15
4.13 theorems	15
4.13.1 Combinatoria	15
4.13.1.1 General	15
4.13.1.2 Triángulo de Pascal	16
4.13.1.3 Números de Catalan	16
4.13.1.4 Números de Stirling (Primera y Segunda Especie)	16
4.13.1.5 Números de Bell	16
4.13.2 Matemáticas	16
4.13.2.1 Fórmulas Generales	16
4.13.2.2 Fórmulas de Vieta	16
4.13.2.3 Sucesión de Fibonacci	16
4.13.2.4 Ternas Pitagóricas	16
4.13.3 Teoría de Números	16
4.13.3.1 Función Totiente de Euler (φ)	...

4.13.3.2 Función e Inversión de Möbius	16
4.13.3.3 MCD (GCD) y MCM (LCM)	16
4.13.4 Operaciones de Bits (Bitwise) y Miscelánea	16
4.13.4.1 Teorema de Erdős-Gallai	17
4.14 xor basis	17
5 Strings	17
5.1 aho corasick	17
5.2 hashing	17
5.3 manacher	17
5.4 prefix function	17
5.5 suffix array	18
5.6 z function	18

1 Data Structure

1.1 2d fenwick tree

```
struct BIT2D {
    vector<vector<int>>> t;
    int n, m;
    BIT2D(int n, int m) : n(n), m(m) {
        t.assign(n, vector<int>(m, 0));
    }
    void update(int x, int y, int add) {
        for (int i = x; i < n; i |= i + 1)
            for (int j = y; j < m; j |= j + 1) t[i][j] += add;
    }
    int sum(int x, int y) {
        int ans = 0;
        for (int i = x; i >= 0; i = (i & (i + 1)) - 1)
            for (int j = y; j >= 0; j = (j & (j + 1)) - 1)
                ans += t[i][j];
        return ans;
    }
    int query(int lx, int rx, int ly, int ry) {
        return sum(rx, ry) - sum(lx - 1, ry) - sum(rx, ly - 1) + sum(lx - 1, ly - 1);
    }
};
```

1.2 centroid decomposition

```
struct Centroid_Decomposition {
    vector<int> adj[N];
    int sz[N], vis[N], aux;
    int pa[N];

    int find(int u, int p, int Sz) {
        int mx = 0, c = 0; sz[u] = 1;
        for (auto v : adj[u]) {
            if (v == p || vis[v]) continue;
            c ^= find(v, u, Sz);
            sz[u] += sz[v];
            mx = max(mx, sz[v]);
        }
        if (max(mx, Sz - sz[u]) <= Sz / 2)
            aux = p, c = u;
        return c;
    }

    void build(int u, int p, int Sz) {
        aux = -1, u = find(u, -1, Sz);
        if (aux != -1) sz[aux] = Sz - sz[u];
        vis[u] = 1, sz[u] = Sz, pa[u] = p;
        for (auto v : adj[u])
            if (!vis[v]) build(v, u, sz[v]);
    }
};
```

```
}
} cd; // by NekoRolly
```

1.3 dynamic segment tree

```
struct nod {
    ll t;
    ll lazy;
    nod *l;
    nod *r;
};

static nod nods[ND];
static int id = 0;
nod *crearnodo() {
    nods[id].t = NEUT, nods[id].lazy = NEUT_LAZY, nods[id].l = nods[id].r = NULL;
    return &nods[id++];
}

ll f(ll a, ll b) {
    return a + b;
}

void aply(ll val, nod *u, ll l, ll r) {
    u->t = (r - l + 1) * val;
    u->lazy = val;
}

struct segmentTree {
    ll n;
    nod *root = NULL;
    segmentTree(ll n) : n(n) {}
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) / 2;
        if (!u->l) u->l = crearnodo();
        if (!u->r) u->r = crearnodo();
        aply(u->lazy, u->l, l, m);
        aply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }

    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (qr < l || r < ql || !u) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
        ll m = (l + r) / 2;
        return f(query(ql, qr, u->l, l, m), query(ql, qr, u->r, m + 1, r));
    }

    ll query(ll ql, ll qr) {
        return query(ql, qr, root, 0, n - 1);
    }

    void update(ll ql, ll qr, ll val, nod *u, ll l, ll r) {
        if (qr < l || r < ql) return;
        if (!u) u = crearnodo();
        if (ql <= l && r <= qr) aply(val, u, l, r);
        else {
            push(u, l, r);
        }
    }
};
```

```
ll m = (l + r) / 2;
update(ql, qr, val, u->l, l, m);
update(ql, qr, val, u->r, m + 1, r);
u->t = f(u->l ? u->l->t : NEUT, u->r ? u->r->t : NEUT);
}
}

void update(ll ql, ll qr, ll val) {
    update(ql, qr, val, root, 0, n - 1);
}

};
```

1.4 fenwick tree

```
struct FenwickTree { // Index 1
    int n, ft[MX + 1];
    void update(int l, int v) {
        for (; l <= n; l += l & -l) ft[l] += v;
    }
    int query(int l) {
        int ans = 0;
        for (; l > 0; l -= l & -l) ans += ft[l];
        return ans;
    }
    int query(int l, int r) { return query(r) - query(l - 1); }
};
```

1.5 hld

```
struct Graph {
    vector<vector<ll>>> adj;
    vector<ll> padre, prof, heavy, head, pos, base, L, R;
    segmentTree st;
    ll curpos, n;
    ll dfs(ll u, ll p) {
        padre[u] = p;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (ll v : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u);
                szu += szv;
                if (mxsz < szv) {
                    mxsz = szv;
                    heavy[u] = v;
                }
            }
        return szu;
    }

    void decompose(ll u, ll h, vector<ll> &a) {
        head[u] = h;
        L[u] = curpos;
    }
};
```

```

    pos[u] = curpos++;
    base[pos[u]] = a[u - 1];
    if (heavy[u] != 0) decompose(heavy[u], h, a);
    for (ll v : adj[u])
        if (v != padre[u] && v != heavy[u]) decompose(v,
v, a);
    R[u] = curpos - 1;
}
void pupdate(ll u, ll val) {
    st.pupdate(pos[u], val);
}
void update(ll u, ll v, ll val) {
    while (head[u] != head[v]) {
        if (prof[head[u]] > prof[head[v]]) swap(u, v);
        st.update(pos[head[v]], pos[v], val);
        v = padre[head[v]];
    }
    if (prof[u] > prof[v]) swap(u, v);
    st.update(pos[u], pos[v], val);
}
ll query(ll u, ll v) {
    ll ans = INF;
    while (head[u] != head[v]) {
        if (prof[head[u]] > prof[head[v]]) swap(u, v);
        ans = f(ans, st.query(pos[head[v]], pos[v]));
        v = padre[head[v]];
    }
    if (prof[u] > prof[v]) swap(u, v);
    ans = f(ans, st.query(pos[u], pos[v]));
    return ans;
}
ll subtreequery(ll u) {
    return st.query(L[u], R[u]);
}
void subtreeupdate(ll u, ll val) {
    st.update(L[u], R[u], val);
}
}
Graph(vector<ll> &a, vector<pll> &aristas, ll root =
1) : n(a.size()) {
    adj.assign(n + 1, vector<ll>()), padre.assign(n + 1,
0), prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [u, v] : aristas) {
        adj[u].pb(v);
        adj[v].pb(u);
    }
    prof[root] = 0;
    dfs(root, root);
    decompose(root, root, a);
    st = segmentTree(base);
}

```

```

    }
};

```

1.6 hld edge

```

struct Graph {
    vector<vector<pll>> adj;
    vector<ll> padre, wpatre, prof, heavy, head, pos, base,
L, R;
    segmentTree st;
    ll curpos;
    ll n;
    ll dfs(ll u, ll p, ll wp) {
        padre[u] = p;
        wpatre[u] = wp;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (auto [v, w] : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u, w);
                szu += szv;
                if (mxsz < szv) mxsz = szv, heavy[u] = v;
            }
        return szu;
    }
    void decompose(ll u, ll h) {
        head[u] = h;
        L[u] = curpos;
        pos[u] = curpos++;
        base[pos[u]] = wpatre[u];
        if (heavy[u] != 0) decompose(heavy[u], h);
        for (auto [v, w] : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v);
        R[u] = curpos - 1;
    }
    void pupdate(ll u, ll v, ll val) {
        if (u == padre[v]) swap(u, v);
        st.pupdate(pos[u], val);
    }
    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        if (u != v) st.update(pos[u] + 1, pos[v], val);
    }
    ll query(ll u, ll v) {
        ll ans = INF;
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
        }
    }
}

```

```

    ans = f(ans, st.query(pos[head[v]], pos[v]));
    v = padre[head[v]];
}
if (prof[u] > prof[v]) swap(u, v);
if (u != v) ans = f(ans, st.query(pos[u] + 1,
pos[v]));
return ans;
}
ll subtreequery(ll u) {
    if (L[u] == R[u]) return 0;
    return st.query(L[u] + 1, R[u]);
}
void subtreeupdate(ll u, ll val) {
    if (L[u] == R[u]) return;
    st.update(L[u] + 1, R[u], val);
}
}
Graph(vector<pair<pll>, ll>> &aristas, ll root = 1) :
n(aristas.size() + 1) {
    adj.assign(n + 1, vector<pll>());
    padre.assign(n + 1, 0), wpatre.assign(n + 1, 0),
prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [edge, w] : aristas) {
        auto [u, v] = edge;
        adj[u].pb({v, w});
        adj[v].pb({u, w});
    }
    prof[root] = 0;
    dfs(root, root, 0LL);
    decompose(root, root);
    st = segmentTree(base);
}
};

```

1.7 merge sort tree

```

struct segmentTree {
    vector<vector<ll>> seg;
    ll n, N;
    segmentTree() {}
    segmentTree(vector<ll> &a) : n(a.size()) {
        N = 1;
        while (N < n) N <= 1;
        seg.assign(2 * N, vector<ll>());
        for (int i = 0; i < n; i++) seg[i + N] = {a[i]};
        for (int i = N - 1; i > 0; i--) {
            auto &l = seg[i < 1], &r = seg[i < 1 | 1];
            seg[i].resize((ll)l.size() + (ll)r.size());
            merge(all(l), all(r), seg[i].begin());
        }
    }
}

```

```
// cantidad de elementos mayores a "a" en el intervalo [ql,qr]
ll query1(ll ql, ll qr, ll a) {
    ll l = ql + N, r = qr + N;
    ll ans = 0;
    while (l <= r) {
        if (l & 1) {
            auto &v = seg[l++];
            ans += v.end() - upper_bound(all(v), a);
        }
        if (!(r & 1)) {
            auto &v = seg[r--];
            ans += v.end() - upper_bound(all(v), a);
        }
        l >>= 1;
        r >>= 1;
    }
    return ans;
}

// cantidad de elementos menores a "a" en el intervalo [ql,qr]
ll query2(ll ql, ll qr, ll a) {
    ll l = ql + N, r = qr + N;
    ll ans = 0;
    while (l <= r) {
        if (l & 1) {
            auto &v = seg[l++];
            ans += lower_bound(all(v), a) - v.begin();
        }
        if (!(r & 1)) {
            auto &v = seg[r--];
            ans += lower_bound(all(v), a) - v.begin();
        }
        l >>= 1;
        r >>= 1;
    }
    return ans;
}
};
```

1.8 multi ordered set

```
// find_by_order(k) -> return iterator to the k-th element (0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num - O(log n)

// using less_equal cause that lower_bound works as
upper_bound and viceversa
// also .find() does not work and you can only erase by
iterator now
// using for example erase(upper_bound(x))
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
template <typename T>
using ordered_set =
    tree<T, null_type, less_equal<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> s;
```

1.9 ordered set

```
// find_by_order(k) -> return iterator to the k-th element (0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num - O(log n)
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
template <typename T>
using ordered_set =
    tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> S;
```

1.10 persistent segment tree

```
const ll MX = 2e5 + 5;
const ll Q = 2e5 + 5;
const ll MXT = 2 * MX + 20 * Q;
struct PST {
    ll root[Q + 1], t[MXT], L[MXT], R[MXT];
    ll n, nodos, lastversion = 0;
    PST(vector<ll> &a) : n(a.size()) {
        nodos = 2 * n - 1;
        root[0] = 1;
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void build(vector<ll> &a, ll id, ll l, ll r) {
        if (l == r) t[id] = a[l];
        else {
            ll m = (l + r) / 2;
            ll idl = id + 1;
            ll idr = id + 2 * (m - l + 1);
            L[id] = idl;
            R[id] = idr;
            build(a, idl, l, m);
            build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(ll ql, ll qr, ll id, ll l, ll r) {
        if (ql <= l && r <= qr) return t[id];
        ll m = (l + r) / 2;
```

```
        if (qr <= m) return query(ql, qr, L[id], l, m);
        if (m + 1 <= ql) return query(ql, qr, R[id], m + 1, r);
        return f(query(ql, qr, L[id], l, m), query(ql, qr, R[id], m + 1, r));
    }
    ll query(ll ql, ll qr, ll version = -1) {
        if (version == -1) version = lastversion;
        return query(ql, qr, root[version], 0, n - 1);
    }
    void update(ll pos, ll val, ll prev, ll &id, ll l, ll r) {
        id = ++nodos;
        if (l == r) t[id] = val;
        else {
            ll m = (l + r) / 2;
            L[id] = L[prev];
            R[id] = R[prev];
            if (pos <= m) update(pos, val, L[prev], L[id], l, m);
            else update(pos, val, R[prev], R[id], m + 1, r);
            t[id] = f(t[L[id]], t[R[id]]);
        }
    }
    void update(ll pos, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        lastversion++;
        update(pos, val, root[version], root[lastversion], 0, n - 1);
    }
};
```

1.11 persistent segment tree lazy

```
const int MX = 2e5 + 5;
const int Q = 2e5 + 5;
const int LOG = 20;
const int ND = 2 * MX + 6 * Q * LOG;
const ll NEUT = 0;
const ll NEUT_LAZY = -2 * INF;
struct nod {
    ll t, lazy;
    nod *l, *r;
};
nod n[ND];
int id = 0;
nod *crearnodo(nod *u = NULL) {
    if (u) n[id].t = u->t, n[id].lazy = u->lazy,
    n[id].l = u->l, n[id].r = u->r;
    else n[id].t = NEUT, n[id].lazy = NEUT_LAZY,
    n[id].l = n[id].r = NULL;
    return &n[id++];
}
ll f(ll a, ll b) {
    return a + b;
```

```

}
void apply(ll val, nod *u, ll l, ll r) {
    u->t += (r - l + 1) * val;
    if (u->lazy == NEUT_LAZY) u->lazy = val;
    else u->lazy += val;
}
struct PST {
    nod *root[MX + 5];
    ll n;
    ll lastversion = -1;
    PST(vector<ll> &a) : n(a.size()) {
        root[++lastversion] = crearnodo();
        build(a, root[0], 0, n - 1);
    }
    void build(vector<ll> &a, nod *u, ll l, ll r) {
        if (!u) u = crearnodo();
        if (l == r) u->t = a[l];
        else {
            ll m = (l + r) >> 1;
            build(a, u->l, l, m);
            build(a, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
    }
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) >> 1;
        u->l = crearnodo(u->l), u->r = crearnodo(u->r);
        apply(u->lazy, u->l, l, m);
        apply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }
    nod *update(ll ql, ll qr, ll val, nod *prev, ll l, ll r) {
        if (qr < l || r < ql) return prev;
        nod *u = crearnodo(prev);
        if (ql <= l && r <= qr) apply(val, u, l, r);
        else {
            push(u, l, r);
            ll m = (l + r) >> 1;
            u->l = update(ql, qr, val, u->l, l, m);
            u->r = update(ql, qr, val, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
        return u;
    }
    void update(ll ql, ll qr, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        root[++lastversion] = update(ql, qr, val,
            root[version], 0, n - 1);
    }
    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (!u || qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
    }
}

```

```

    ll m = (l + r) >> 1;
    return f(query(ql, qr, u->l, l, m), query(ql, qr, u->r, m + 1, r));
}
ll query(ll ql, ll qr, ll version = -1) {
    if (version == -1) version = lastversion;
    return query(ql, qr, root[version], 0, n - 1);
}
};

```

1.12 segment tree

```

struct IterativeSegmentTree { // Index 0
    int n;
    ll ST[2 * MX];
    const ll NEUT = 0;
    void build(vector<ll> &a) {
        n = a.size();
        for (int i = n; i < (n << 1); i++) ST[i] = a[i - n];
        for (int i = n - 1; i > 0; i--) ST[i] = f(ST[i << 1], ST[i << 1 | 1]);
    }
    ll f(ll x, ll y) { return max(x, y); }
    void update(int i, ll val) {
        for (ST[i += n] = val; i >= 1; i >>= 1) ST[i] = f(ST[i << 1], ST[i << 1 | 1]);
    }
    ll query(int l, int r) { // [l, r)
        ll ansL = NEUT, ansR = NEUT;
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) ansL = f(ansL, ST[l++]);
            if (r & 1) ansR = f(ST[--r], ansR);
        }
        return f(ansL, ansR);
    }
};

```

1.13 segment tree 2d

```

struct ST2D { //Index 0
    int n, m;
    vector<vector<ll>> t;
    ll f(ll a, ll b) {
        return a + b;
    }
    ST2D(vector<vector<ll>> &a) {
        n = a.size();
        m = a[0].size();
        t.assign(2 * n, vector<ll>(2 * m, 0));
        for (int i = 0; i < n; i++)
            for (int j = 0; j < m; j++) t[i + n][j + m] = a[i][j];
        for (int i = n; i < 2 * n; i++)
            for (int j = m - 1; j > 0; j--) t[i][j] = f(t[i], t[i << 1], t[i << 1 | 1]);
    }
};

```

```

    for (int i = n - 1; i > 0; i--)
        for (int j = 1; j < 2 * m; j++) t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
}
void update(int x, int y, ll val) {
    x += n, y += m, t[x][y] = val;
    for (int j = y; j > 1; j >>= 1) t[x][j >> 1] = f(t[x][j], t[x][j ^ 1]);
    for (int i = x >> 1; i > 0; i >>= 1) {
        int j = y;
        t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
        for (; j > 1; j >>= 1) t[i][j >> 1] = f(t[i][j], t[i][j ^ 1]);
    }
}
ll queryY(int i, int l, int r) {
    ll ans = 0;
    for (l += m, r += m + 1; l < r; l >>= 1, r >>= 1) {
        if (l & 1) ans = f(ans, t[l][l++]);
        if (r & 1) ans = f(t[r][--r], ans);
    }
    return ans;
}
ll query(int lx, int rx, int ly, int ry) {
    ll ans = 0;
    for (x1 += n, x2 += n + 1; x1 < x2; x1 >>= 1, x2 >>= 1) {
        if (x1 & 1) ans = f(ans, queryY(x1++, y1, y2));
        if (x2 & 1) ans = f(queryY(--x2, y1, y2), ans);
    }
    return ans;
}
};

```

1.14 segment tree beats

```

struct node {
    ll sum;
    ll max1, max2, cont; // min para chmax
};
node f(node a, node b) { // simetrico con min para chmax
    node ans;
    ans.sum = a.sum + b.sum;
    ans.max1 = max(a.max1, b.max1);
    if (a.max1 == b.max1) ans.max2 = max(a.max2, b.max2),
    ans.cont = a.cont + b.cont;
    else if (a.max1 > b.max1) ans.max2 = max(a.max2, b.max1), ans.cont = a.cont;
    else ans.max2 = max(a.max1, b.max2), ans.cont = b.cont;
    return ans;
}
struct SegmentTree {
    vector<node> t;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {
    }
};

```

```

t.resize(2 * n);
build(a, 1, 0, n - 1);
}
void build(vector<ll> &a, int id, int l, int r) {
    if (l == r) t[id] = {a[l], a[l], -INF, 1}; // chmax:
-INF -> INF
    else {
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        build(a, idl, l, m), build(a, idr, m + 1, r);
        t[id] = f(t[idl], t[idr]);
    }
}
void apply(ll val, int id) {
    if (t[id].max1 <= val) return; // chmax: (<=) ->
(>=)
    t[id].sum += t[id].cont * (val - t[id].max1);
    t[id].max1 = val;
}
void push(int id, int l, int r) {
    if (l == r) return;
    int m = (l + r) >> 1;
    int idl = id + 1;
    int idr = id + 2 * (m - l + 1);
    apply(t[id].max1, idl);
    apply(t[id].max1, idr);
}
void update(int ql, int qr, ll val, int id, int l, int
r) {
    if (qr < l || r < ql || t[id].max1 <= val)
return; //chmax: (<=) -> (>=)
    if (ql <= l && r <= qr && t[id].max2 < val)
apply(val, id); // chmax: (<) -> (>)
    else {
        push(id, l, r);
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
        t[id] = f(t[idl], t[idr]);
    }
}
ll query(int ql, int qr, int id, int l, int r) {
    if (qr < l || r < ql) return NEUT;
    if (ql <= l && r <= qr) return t[id].sum;
    push(id, l, r);
    int m = (l + r) >> 1;
    int idl = id + 1;
    int idr = id + 2 * (m - l + 1);
    return query(ql, qr, idl, l, m) + query(ql, qr, idr,
m + 1, r);
}
};

```

1.15 segment tree lazy

```

struct SegmentTree { // id: index 1. Vector a: Index 0
    vector<ll> t, lazy;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {
        t.assign(2 * n, NEUT), lazy.assign(2 * n,
NEUT_LAZY);
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void aply(ll val, int id, int l, int r) {
        t[id] += (r - l + 1) * val;
        lazy[id] += val;
    }
    void build(vector<ll> &a, int id, int l, int r) {
        if (l == r) t[id] = a[l];
        else {
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            build(a, idl, l, m), build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    void push(int id, int l, int r) {
        if (l == r || lazy[id] == NEUT_LAZY) return;
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        aply(lazy[id], idl, l, m);
        aply(lazy[id], idr, m + 1, r);
        lazy[id] = NEUT_LAZY;
    }
    void update(int ql, int qr, ll val, int id, int l, int
r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) aply(val, id, l, r);
        else {
            push(id, l, r);
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(int ql, int qr, int id, int l, int r) {
        if (qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return t[id];
        push(id, l, r);
        int m = (l + r) >> 1;
    }
};

```

```

int idl = id + 1;
int idr = id + 2 * (m - l + 1);
return f(query(ql, qr, idl, l, m), query(ql, qr,
idr, m + 1, r));
}
};

```

1.16 sparse table

```

const int LG = __lg(MX) + 1;
int st[MX][LG];
void build(vector<int> &A) { // O(|f| n log n)
    int n = A.size();
    for (int i = 0; i < n; i++) st[i][0] = A[i];
    for (int k = 1; k < LG; k++) {
        for (int i = 0; i + (1 << k) <= n; i++) {
            st[i][k] = min(st[i][k - 1], st[i + (1 << (k -
1))][k - 1]);
        }
    }
}
int query(int l, int r) { // O(|f|)
    int lg = __lg(r - l + 1);
    return min(st[l][lg], st[r - (1 << lg) + 1][lg]);
}

```

1.17 trie bitmask

```

struct Trie{
    int t[N*B][2], sz[N*B];
    int cur;

    void insert(int x){
        int u = 0; sz[u]++;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(int x){
        int u = 0; sz[u]--;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            u = t[u][k], sz[u]--;
        }
    }

    int get_min(int x){ // min xor
        int u = 0, ans = 0;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] != 0 && sz[t[u][k]] > 0) u = t[u]
[k];
        }
    }
};

```

```

        else u = t[u][k^1], ans ^= 1<<i;
    }
    return ans;
}

int get_max(int x){ // max xor
    int u = 0, ans = 0;
    for (int i=B-1; i>=0; i--){
        int k = x>>i&1;
        if (t[u][k^1] != 0 && sz[t[u][k^1]] > 0)
            u = t[u][k^1], ans ^= 1<<i;
        else u = t[u][k];
    }
    return ans;
}
} tr; // by NekoRolly

```

1.18 trie string

```

struct Trie{
    int t[N][K], sz[N], cur;

    void insert(string &s){
        int u = 0; sz[u]++;
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(string &s){
        int u = 0; sz[u]--;
        for (char &c : s){
            int k = c-'a';
            u = t[u][k], sz[u]--;
        }
    }
} tr; // by NekoRolly

```

2 Geometry

2.1 convex hull trick

```

struct Line{
    ll m, b;

    ll y(ll x){ return m*x+b; }
};

struct Convex_Hull_Track{
    vector<Line> vec;
    int l = 0;

```

```

bool check(int i){
    double x1 = double(vec[i-1].b - vec[i+1].b)/
        (vec[i+1].m - vec[i-1].m);
    double x2 = double(vec[i].b - vec[i+1].b)/
        (vec[i+1].m - vec[i].m);
    return x1 >= x2;
}

void add(Line L){
    vec.push_back(L);
    while (vec.size() >= 3 && check(vec.size()-2))
        vec.erase(vec.end()-2);
    l = min(l, int(vec.size()-1));
}

ll query(ll x){ // O(1) amortizado
    while (l+1 < vec.size() && vec[l].y(x) >
        vec[l+1].y(x)) l++;
    return vec[l].y(x);
}

ll find(ll x){ // O(logn)
    int l = 0, r = vec.size();
    while (l+1 < r){
        int m = (l+r)>>1;
        if (vec[m-1].y(x) > vec[m].y(x)) l = m;
        else r = m;
    }
    return vec[l].y(x);
}
} cht; // by NekoRolly

```

2.2 li chao tree

```

struct Line{
    ll m, b;

    ll y(ll x){ return m*x+b; }
};

struct Li_Chao_Tree{
    Line t[N*4];

    void update(int id, int l, int r, Line L){
        if (l+1 == r || t[id].b == -infll){
            if (L.y(l) > t[id].y(l)) t[id] = L;
            return;
        }
        int m = (l+r)>>1;
        if (t[id].y(m) < L.y(m)) swap(t[id], L);
        if (t[id].m > L.m) update(id<<1, l, m, L);
        else update(id<<1|1, m, r, L);
    }

    ll query(int id, int l, int r, ll x){

```

```

        if (l+1 == r) return t[id].y(x);
        int m = (l+r)>>1;
        if (x < m) return max(t[id].y(x), query(id<<1, l, m,
            x));
        else return max(t[id].y(x), query(id<<1|1, m, r,
            x));
    }
} lct; // by NekoRolly

```

3 Graphs

3.1 2sat

```

struct SAT { // Index_1
    vector<vector<int>> adj, adjt;
    vector<bool> used;
    vector<int> comp;
    int n;
    stack<int> st;
    SAT(int n) : n(n) {
        adj.resize(2 * n + 1), adjt.resize(2 * n + 1);
        used.assign(2 * n + 1, 0), comp.assign(2 * n + 1,
            0);
    }

    void add(int p, bool bp, int q, bool bq) { // add p -> q
        p += (!bp) * n;
        q += (!bq) * n;
        adj[p].pb(q);
        adjt[q].pb(p);
    }

    void addor(int p, bool bp, int q, bool bq) { // add p or q
        add(p, !bp, q, bq);
        add(q, !bq, p, bp);
    }

    void addxor(int p, bool bp, int q, bool bq) { // add p
        addor(p, bp, q, bq);
        addor(p, !bp, q, !bq);
    }

    void addnxor(int p, bool bp, int q, bool bq) { // add
        ~((p xor q)
            addor(p, !bp, q, bq);
            addor(p, bp, q, !bq);
        )
    }

    void dfs(int u) {
        used[u] = 1;
        for (int v : adj[u])
            if (!used[v]) dfs(v);
        st.push(u);
    }

    void _dfs(int u, int c) {
        comp[u] = c;
        for (int v : adjt[u])

```



```

        if (!comp[v]) _dfs(v, c);
    }
    bool solve(vector<bool> &ans) {
        ans.resize(n + 1);
        for (int u = 1; u <= 2 * n; u++)
            if (!used[u]) dfs(u);
        int c = 0;
        while (!st.empty()) {
            int u = st.top();
            st.pop();
            if (!comp[u]) _dfs(u, ++c);
        }
        for (int p = 1; p <= n; p++) {
            if (comp[p] == comp[p + n]) return false;
            ans[p] = comp[p] > comp[p + n];
        }
        return true;
    }
};

```

3.2 bellman ford

```

// find distance from source node to all nodes.
// supports negative edge weights.
// returns true if a negative cycle is detected.
bool bellmanFord(int s, int n) { // O(VE)
    d.assign(n + 1, INF);
    vector<bool> cycle(n, false);
    bool f = false;
    for (int i = 0; i < n; i++) {
        f = false;
        for (int u = 0; u < n; u++) {
            for (auto [v, w] : adj[u]) {
                if (d[u] != INF && d[u] + w < d[v]) {
                    d[v] = d[u] + w;
                    parent[v] = u;
                    f = true;
                    if (i == n - 1) cycle[v] = true;
                }
            }
        }
        if (!f) return false;
    }
    return true;
}

```

3.3 blossom

```

struct Graph {
    vector<int> adj[MX];
    int p[MX];
    int match[MX];
    // match[u] is the node match with u
    // or -1 if it does not belong to the matching

```

```

    int base[MX]; // base of the flower
    bool used[MX];
    bool blossom[MX];

    void addEdge(int u, int v);

    int lca(int u, int v, int n) { // O(n)
        vector<bool> marked(n);
        while (true) {
            u = base[u]; // blossom compression
            marked[u] = true;
            if (match[u] == -1) break; // we reach the
            root : unexposed vertex
            u = p[match[u]];
        }
        while (true) {
            v = base[v];
            if (marked[v]) return v;
            v = p[match[v]];
        }
        return -1;
    }

    void markPath(int u, int b, int children) {
        while (base[u] != b) {
            blossom[base[u]] = blossom[base[match[u]]] =
true;
            p[u] = children, children = match[u], u =
p[match[u]];
        }
    }

    void compress(int u, int v, int n, queue<int> &q) {
        int curBase = lca(u, v, n); // base of the flower
        assert(curBase != -1);
        fill(blossom, blossom + n, false);

        markPath(u, curBase, v);
        markPath(v, curBase, u);
        for (int i = 0; i < n; i++) {
            if (blossom[base[i]]) {
                base[i] = curBase;
                if (!used[i]) used[i] = true, q.push(i);
            }
        }
    }

    int findPath(int root, int n) {
        for (int i = 0; i < n; i++) base[i] = i, used[i] =
false, p[i] = -1;
        used[root] = true;
        queue<int> q;
        q.push(root);
        while (!q.empty()) {
            int u = q.front();

```

```

            q.pop();
            for (int to : adj[u]) {
                if (base[u] == base[to] || match[u] == to)
continue;
                if (to == root || (match[to] != -1 &&
p[match[to]] != -1))
                    compress(u, to, n, q); // we find a
blossom
                else if (p[to] == -1) {
                    // augmenting path or continue bfs
                    p[to] = u;
                    if (match[to] == -1) return to; // we
find an augmenting path
                    // we continue the bfs
                    to = match[to], used[to] = true,
q.push(to);
                }
            }
            return -1;
        }

        int maxMatching(int n) { // O(n^3)
            fill(match, match + n, -1);
            int ans = 0;
            for (int u = 0; u < n; u++) {
                if (match[u] == -1) {
                    int v = findPath(u, n);
                    if (v != -1) {
                        ans++;
                        while (v != -1) {
                            int nextVertex = match[p[v]];
                            match[v] = p[v], match[p[v]] = v, v
= nextVertex;
                        }
                    }
                    return ans;
                }
            }
        }
    } G;

```

3.4 dinic

```

// This implementation is for small graphs G(V, E) (|V| <=
3000 aprox)
// where O(|V|^2) memory will fit the memory limit.
using Cap = long long;
const int MX = 3000;
const Cap INF = 1e18;
struct Graph {
    vector<int> adj[MX];
    Cap cap[MX][MX];
    Cap flow[MX][MX];
    bool added[MX][MX];

```



```

int level[MX];
int nextEdge[MX];

void clear(int n) {
    for (int i = 0; i < n; i++) {
        adj[i].clear();
        for (int j = 0; j < n; j++) cap[i][j] = 0,
        flow[i][j] = 0, added[i][j] = false;
    }
}

void addEdge(int u, int v, Cap w, bool dir) {
    if (!added[u][v]) {
        adj[u].push_back(v);
        adj[v].push_back(u);
        added[u][v] = added[v][u] = true;
    }
    cap[u][v] += w;
    if (!dir) cap[v][u] += w;
}

Cap dfs(int u, int t, Cap f) { // O(E)
    if (u == t) return f;
    for (int &i = nextEdge[u]; i < adj[u].size(); i++) {
        int v = adj[u][i];
        Cap cf = cap[u][v] - flow[u][v];
        if (cf == 0 || level[v] != level[u] + 1)
            continue;
        Cap ret = dfs(v, t, min(f, cf));
        if (ret > 0) {
            flow[u][v] += ret;
            flow[v][u] -= ret;
            return ret;
        }
    }
    return 0;
}

bool bfs(int s, int t, int n) { // O(E)
    fill(level, level + n, -1);
    queue<int> q({s});
    level[s] = 0;
    while (!q.empty()) {
        int u = q.front();
        nextEdge[u] = 0;
        q.pop();
        for (int v : adj[u]) {
            Cap cf = cap[u][v] - flow[u][v];
            if (level[v] == -1 && cf > 0) {
                level[v] = level[u] + 1;
                q.push(v);
            }
        }
    }
    return level[t] != -1;
}

```

```

}

Cap maxFlow(int s, int t, int n) {
    // General: O(E * V^2), O(E * V + V * |F|)
    // Unit Cap: O(E * min(E^(1/2), V^(2/3)))
    // Unit Network: O(E * V^(1/2))
    // In unit network, all the edges have unit capacity
    // and for any vertex except s and t either the
    // incoming or outgoing edge is unique.
    Cap ans = 0;
    while (bfs(s, t, n)) { // O(V) iterations
        // after bfs, the graph is a DAG
        while (Cap inc = dfs(s, t, INF)) ans += inc;
    }
}

} G;

```

3.5 dominator tree

```

//idom[i]=parent of i in dominator tree with root=rt, or -1
//if not exists
int
n, rnk[MXN], pre[MXN], anc[MXN], idom[MXN], semi[MXN], low[MXN];
vector<int> g[MXN], rev[MXN], dom[MXN], ord;
void dfspre(int pos) {
    rnk[pos] = SZ(ord); ord.pb(pos);
    for (auto x : g[pos]) {
        rev[x].pb(pos);
        if (rnk[x] == n) pre[x] = pos, dfspre(x);
    }
}

int eval(int v) {
    if (anc[v] < n && anc[anc[v]] < n) {
        int x = eval(anc[v]);
        if (rnk[semi[low[v]]] > rnk[semi[x]]) low[v] = x;
        anc[v] = anc[anc[v]];
    }
    return low[v];
}

void dominators(int rt) {
    for (int i, n) {
        dom[i].clear(); rev[i].clear();
        rnk[i] = pre[i] = anc[i] = idom[i] = n;
        semi[i] = low[i] = i;
    }
    ord.clear(); dfspre(rt);
    for (int i = SZ(ord) - 1; i--;) {
        int w = ord[i];
        for (int v : rev[w]) {
            int u = eval(v);
            if (rnk[semi[w]] > rnk[semi[u]]) semi[w] = semi[u];
        }
        dom[semi[w]].pb(w); anc[w] = pre[w];
        for (int v : dom[pre[w]]) {

```

```

            int u = eval(v);
            idom[v] = (rnk[pre[w]] > rnk[semi[u]] ? u : pre[w]);
        }
        dom[pre[w]].clear();
    }
    for (int w : ord) if (w != rt && idom[w] != semi[w])
        idom[w] = idom[idom[w]];
    for (int i = 0; i < n; i++) if (idom[i] == n) idom[i] = -1;
}

```

3.6 floyd warshall

```

void Ini(int n) {
    for (int u, n) {
        for (int v, n) d[u][v] = INF, parent[u][v] = -1;
        d[u][u] = 0;
    }
}

void addEdge(int u, int v, int w) {
    if (w < d[u][v]) {
        d[u][v] = w;
        parent[u][v] = u;
    }
}

void floydWarshall(int n) { // O(V^3)
    for (int k, n) for (int u, n) for (int v, n) {
        if (d[u][k] == INF) continue;
        if (d[k][v] == INF) continue;
        if (d[u][k] + d[k][v] < d[u][v]) {
            d[u][v] = max(-INF, d[u][k] + d[k][v]);
            parent[u][v] = parent[k][v];
        }
    }

    // negative cycles
    for (int u, n) for (int v, n) for (int k, n) {
        if (d[k][k] < 0 && d[u][k] != INF && d[k][v] != INF)
        {
            d[u][v] = -INF;
        }
    }
}

```

3.7 kuhn

```

// Maximum matching for bipartite unweighted graphs
struct Graph {
    int match[MX];
    bool vis[MX];
    vector<int> adj[MX];
    void addEdge(int u, int v);

    bool dfs(int u) {

```

```

    if (vis[u]) return false;
    vis[u] = true;
    for (int v : adj[u]) {
        if (match[v] == -1 || dfs(match[v])) {
            match[v] = u, match[u] = v;
            return true;
        }
    }
    return false;
}

int maximumMatching(int n) { // O(V * E)
    // If the bipartition is (V1, V2) we can iterate
    // from the side that have less nodes (say is V1) and achieve O(V1 * E)
    fill(match, match + n, -1);
    int ans = 0;
    for (int u = 0; u < n; u++) {
        if (match[u] == -1) {
            fill(vis, vis + n, false);
            ans += dfs(u);
        }
    }
    return ans;
}
} G;

```

3.8 lca

```

int tIn[MX], tOut[MX];
int anc[MX][LG];
int timer;

void dfs(int u) {
    tIn[u] = timer++;
    for (int j = 1; j < LG; j++) anc[u][j] = anc[anc[u][j - 1]][j - 1];
    for (int v : adj[u]) {
        if (v == anc[u][0]) continue;
        anc[v][0] = u;
        dfs(v);
    }
    tOut[u] = timer++;
}

bool isAnc(int u, int v) { // is u ancestor of v ?
    return tIn[u] <= tIn[v] && tOut[u] >= tOut[v];
}

int lca(int u, int v) { // O(log n)
    if (isAnc(u, v)) return u;
    if (isAnc(v, u)) return v;
    for (int i = LG - 1; i >= 0; i--) {
        if (!isAnc(anc[u][i], v)) u = anc[u][i];
    }
}

```

```

    }
    return anc[u][0];
}

```

3.9 maxflow mincost

```

using Cap = long long;
using Cost = long long;

const int MX = 5000;
const Cap INF_CAP = 1e18;
const Cost INF_COST = 1e18;

struct Edge {
    int to;
    Cap flow, cap;
    Cost cost;
    int rev; // index of the backward edge in the adj list
    Edge(int to, Cap cap, Cost cost, int rev)
        : to(to), flow(0), cap(cap), cost(cost), rev(rev) {}
};

struct Graph {
    vector<Edge> adj[MX];
    int parentEdge[MX];
    Cost pot[MX];
    //|pot[u]| <= the maximum sum of absolute cost in a path
    // which is also bounded by [(V - 1) * C]
    // where C is the maximum value of |cost(e)| for all
    // edges e
    void clear(int n) {
        for (int i = 0; i < n; i++) {
            adj[i].clear();
            pot[i] = 0;
        }
    }

    void addEdge(int u, int v, Cap w, Cost cost, bool dir) {
        adj[u].push_back(Edge(v, w, cost, adj[v].size()));
        adj[v].push_back(Edge(u, 0, -cost, adj[u].size() - 1));
        if (!dir) addEdge(v, u, w, cost, true);
    }

    void spfa(int s, int n) { // O(E V)
        vector<bool> inQueue(n, false);
        for (int i = 0; i < n; i++) pot[i] = INF_COST;
        queue<int> q;
        pot[s] = 0;
        inQueue[s] = true;
        q.push(s);
        while (!q.empty()) {
            int u = q.front();
            q.pop();

```

```

            inQueue[u] = false;
            for (Edge e : adj[u]) {
                int v = e.to;
                if (e.cap - e.flow > 0 && pot[u] + e.cost <
                    pot[v]) {
                    pot[v] = pot[u] + e.cost;
                    if (!inQueue[v]) q.push(v);
                    inQueue[v] = true;
                }
            }
        }

        void pushFlow(int s, int t, Cap inc) {
            int v = t;
            while (v != s) {
                Edge &backward = adj[v][parentEdge[v]];
                int u = backward.to;
                Edge &forward = adj[u][backward.rev];
                forward.flow += inc;
                backward.flow -= inc;
                v = u;
            }
        }

        pair<Cap, Cost> dijkstra(int s, int t, int n) { // O(E log V)
            //<flow, cost>
            using Path = pair<Cost, int>; //<weight, node>
            priority_queue<Path, vector<Path>, greater<Path>> q;
            vector<Cost> d(n, INF_COST);
            vector<Cap> residualCap(n, 0);
            d[s] = 0;
            residualCap[s] = INF_CAP;
            q.push(Path(d[s], s));
            while (!q.empty()) {
                auto [dist, u] = q.top();
                q.pop();
                if (dist != d[u]) continue;
                for (Edge e : adj[u]) {
                    int v = e.to;
                    Cap cf = e.cap - e.flow;
                    Cost cost = e.cost + pot[u] - pot[v];
                    if (cf > 0 && d[u] + cost < d[v]) {
                        d[v] = d[u] + cost;
                        q.push(Path(d[v], v));
                        residualCap[v] = min(residualCap[v], cf);
                        parentEdge[v] = e.rev;
                    }
                }
            }
            if (d[t] == INF_COST) return {0, 0};
            for (int i = 0; i < n; i++) {
                if (pot[i] != INF_COST) pot[i] += d[i];
            }
        }
    }
}

```

```

    }
    Cap cf = residualCap[t];
    pushFlow(s, t, cf);
    return {cf, pot[t] * cf};
}

pair<Cap, Cost> minCostFlow(int s, int t, int n) {
    // O(E log V * maxFlow)
    // maxFlow <= V * U, where U is the maximum capacity
    // Assumption: Initially no negative cycles
    // <maxFlow, minCost>
    spfa(s, n); // not necessary if there is no negative
edges
    pair<Cap, Cost> inc;
    Cap flow = 0;
    Cost cost = 0;
    do {
        inc = dijkstra(s, t, n);
        flow += inc.first;
        cost += inc.second;
    } while (inc.first > 0);
    return {flow, cost};
}
} G;

```

3.10 tree bridge

```

vector<int> adj[MX];
int tin[MX], low[MX], timer;
vector<pii> bridges;
int component[MX];
vector<int> nods[MX];
vector<array<int, 3>> adjcomp[MX];
void dfs(int u, int p) {
    tin[u] = low[u] = ++timer;
    for (int v : adj[u])
        if (v != p) {
            if (tin[v]) low[u] = min(low[u], tin[v]);
            else dfs(v, u), low[u] = min(low[u], low[v]);
            if (tin[u] < low[v]) bridges.pb({u, v});
        }
}
bool isbridge(int u, int v) {
    if (tin[u] > tin[v]) swap(u, v);
    return tin[u] < low[v];
}
void dfsc(int u, int comp) {
    component[u] = comp;
    nods[comp].pb(u);
    for (int v : adj[u])
        if (!isbridge(u, v) && !component[v]) dfsc(v, comp);
}
int tree_bridge(int n) {
    dfs(1, 1);
    int comp = 0;
}

```

```

for (int u = 1; u <= n; u++)
    if (!component[u]) dfsc(u, ++comp);
for (auto [u, v] : bridges) {
    int cu = component[u], cv = component[v];
    adjcomp[cu].pb({cv, u, v});
    adjcomp[cv].pb({cu, u, v});
}
return comp;
}

```

4 Math

4.1 cht

```

struct Line {
    mutable ll a, b, c;

    bool operator<(Line r) const {
        return a < r.a;
    }
    bool operator<(ll x) const {
        return c < x;
    }
};

// dynamically insert `a*x + b` lines and query for maximum
// at any x all operations have complexity O(log N)
struct LineContainer : multiset<Line, less<>> {
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    bool isect(iterator x, iterator y) {
        if (y == end()) return x->c = INF, 0;
        if (x->a == y->a) x->c = x->b > y->b ? INF : -INF;
        else x->c = div(y->b - x->b, x->a - y->a);
        return x->c >= y->c;
    }

    void add(ll a, ll b) {
        // a *= -1, b *= -1 // for min
        auto z = insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->c >= y->c)
            isect(x, erase(y));
    }

    ll query(ll x) {
        if (empty()) return -INF; // INF for min
        auto l = *lower_bound(x);
        return l.a * x + l.b;
        // return -l.a * x - l.b; // for min
    }
}

```

```

}
};

```

4.2 criba

```

bool criba[MX];
ll primerfactor[MX], phi[MX];
// criba para primerfactor, phi O(nlog(log(n)))
void completar() {
    for (int i = 0; i < MX; i++) {
        criba[i] = 1;
        primerfactor[i] = i;
        phi[i] = i;
    }
    criba[0] = 0;
    criba[1] = 0;
    for (ll i = 2; i < MX; i++) {
        if (criba[i]) {
            phi[i]--;
            for (ll j = 2 * i; j < MX; j += i) {
                criba[j] = 0;
                phi[j] -= phi[j] / i;
                if (primerfactor[j] == j) primerfactor[j] = i;
            }
        }
    }
}

vector<pll> factorizacioncriba(ll n) { // O(log(n))
    vector<pll> ans;
    while (n > 1) {
        ll f = primerfactor[n];
        ll pot = 0;
        while (primerfactor[n] == f) {
            n /= f;
            pot++;
        }
        ans.pb({f, pot});
    }
    return ans;
}

vector<ll> divisorescriba(ll n) { // O(d(x))
    vector<ll> fact = {1};
    while (n > 1) {
        ll f = primerfactor[n];
        ll sz = fact.size();
        ll divisor = 1;
        while (n % f == 0) {
            divisor *= f;
            n /= f;
            rep(i, sz) fact.pb(fact[i] * divisor);
        }
    }
    return fact;
}

```

4.3 diophantine

```
ll extended_gcd(ll a, ll b, ll &x, ll &y) {
    x = 1, y = 0;
    ll x1 = 0, y1 = 1;
    while (b) {
        ll q = a / b;
        tie(x, x1) = make_pair(x1, x - q * x1);
        tie(y, y1) = make_pair(y1, y - q * y1);
        tie(a, b) = make_pair(b, a - q * b);
    }
    return a < 0 ? (x *= -1, y *= -1, a *= -1, a) : a;
}

bool diophantine(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (a == 0 && b == 0) return c ? 0 : (x = y = 0, 1);
    g = extended_gcd(a, b, x, y);
    return (c % g == 0 ? (x *= c / g, y *= c / g, 1) : 0);
}

ll divdown(ll a, ll b) {
    return a / b - ((a ^ b) < 0 && a % b);
}

ll divup(ll a, ll b) {
    return a / b + ((a ^ b) >= 0 && a % b);
}

vector<pll> allsolution(ll a, ll b, ll c, ll lx, ll rx, ll ly, ll ry) {
    // x = x0 + k*(b/g)    y = y0 - k*(a/g)
    ll x0, y0, g;
    if (!diophantine(a, b, c, x0, y0, g)) return {};
    vector<pll> ans;
    if (!g) {
        for (ll x = lx; x <= rx; x++)
            for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    if (!a) {
        ll y = c / b;
        if (!(ly <= y && y <= ry)) return {};
        for (ll x = lx; x <= rx; x++) ans.pb({x, y});
        return ans;
    }
    if (!b) {
        ll x = c / a;
        if (!(lx <= x && x <= rx)) return {};
        for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    ll dx = b / g, dy = a / g;
    ll kmin, kmax;
    if (dx > 0) kmin = divup(lx - x0, dx), kmax = divdown(rx - x0, dx);
    else kmin = divup(rx - x0, dx), kmax = divdown(lx - x0, dx);
    if (dy > 0) kmin = max(kmin, divup(y0 - ry, dy)), kmax = min(kmax, divdown(y0 - ly, dy));
}
```

```
    else kmin = max(kmin, divup(y0 - ly, dy)), kmax = min(kmax, divdown(y0 - ry, dy));
    for (ll k = kmin; k <= kmax; k++) ans.pb({x0 + k * dx, y0 - k * dy});
    return ans;
}

bool positive_sol_minf(ll a, ll b, ll c, ll &x, ll &y, ll cx, ll cy) {
    // f(x,y) = cx*x + cy*y
    ll g;
    if (!diophantine(a, b, c, x, y, g)) return 0;
    ll dx = b / g, dy = a / g;
    ll kmin = -INF, kmax = INF;
    dx > 0 ? kmin = max(kmin, divup(-x, dx)) : kmax = min(kmax, divdown(-x, dx));
    dy < 0 ? kmin = max(kmin, divup(y, dy)) : kmax = min(kmax, divdown(y, dy));
    if (kmin > kmax) return 0;
    ll df = cx * dx - cy * dy;
    ll k = df >= 0 ? kmin : kmax; // df < 0 para max
    x += dx * k;
    y -= dy * k;
    return 1;
}
```

4.4 fft

```
typedef complex<double> cd;
vector<vector<cd>> w;
void calc(int log) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <= 1) {
        w[l].resize(len / 2);
        cd wn(cos(2 * PI / len), sin(2 * PI / len));
        for (int j = 0; 2 * j < len; j++)
            w[l][j] = (j & 1 ? w[l - 1][j / 2] * wn : w[l - 1][j / 2]);
    }
}

void fft(vector<cd> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                cd u = a[i + j], v = a[i + j + len / 2] *
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
            }
}
```

```
    }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        for (int i = 0; i < n; i++) a[i] /= n;
    }
}

vector<ll> polymult(vector<ll> &a, vector<ll> &b) {
    int n = 1, log = 0;
    while (n < a.size() + b.size() - 1) n <= 1, log++;
    calc(log);
    vector<cd> A(n);
    for (int i = 0; i < n; i++) A[i] = cd(i < a.size() ? a[i] : 0, i < b.size() ? b[i] : 0);
    fft(A, 0);
    for (int i = 0; i < n; i++) A[i] *= A[i];
    fft(A, 1);
    vector<ll> ans(a.size() + b.size() - 1);
    for (int i = 0; i < ans.size(); i++) ans[i] = round(A[i].imag() / 2);
    return ans;
}

// 2D-FFT
void fft2D(vector<vector<cd>> &a, bool invert) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) fft(a[i], invert);
    vector<cd> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        fft(b, invert);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}

vector<vector<ll>> convolution2D(vector<vector<ll>> &a, vector<vector<ll>> &b) {
    int n = 1, logn = 0;
    while (n < a.size() + b.size() - 1) n <= 1, logn++;
    int m = 1, logm = 0;
    while (m < a[0].size() + b[0].size() - 1) m <= 1, logm++;
    calc(max(logn, logm));
    vector<vector<cd>> A(n, vector<cd>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            A[i][j] = cd(i < a.size() && j < a[0].size() ? a[i][j] : 0, i < b.size() && j < b[0].size() ? b[i][j] : 0);
    fft2D(A, 0);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) A[i][j] *= A[i][j];
    fft2D(A, 1);
    vector<vector<ll>> ans(a.size() + b.size() - 1, vector<ll>(a[0].size() + b[0].size() - 1));
    for (int i = 0; i < ans.size(); i++)
        for (int j = 0; j < ans[0].size(); j++) ans[i][j] =
}
```

```
round(A[i][j].imag() / 2);
return ans;
}
```

4.5 fwht bitwiseconvolution

```
//OR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j + h] +=
a[j];
}
void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = s >> 1; s >= 2; s >= 1, h >= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j + h] -=
A[j];
}
vector<ll> OrConvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
vector<ll> SubsetConvolution(vector<ll> a, vector<ll> b) {
    int n = 1, bit = 1;
    while (n < a.size() || n < b.size()) n <= 1, bit++;
    a.resize(n), b.resize(n);
    vector<vector<ll>> A(bit + 1, vector<ll>(n)), B(bit + 1,
vector<ll>(n));
    for (int i = 0; i < n; i++) {
        int f = __builtin_popcount(i);
        A[f][i] = a[i], B[f][i] = b[i];
    }
    for (int i = 0; i <= bit; i++) fwht(A[i]), fwht(B[i]);
    vector<ll> conv(n, 0);
    for (int k = 0; k < n; k++) {
        int f = __builtin_popcount(k);
        for (int i = 0; i <= f; i++) conv[k] += A[i][k] *
B[f - i][k];
    }
    ifwht(conv);
    return conv;
}
//AND-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
```

```
    for (int i = 0; i < n; i += s)
        for (int j = i; j < i + h; j++) a[j] += a[j +
h];
}
void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = n >> 1; s >= 2; s >= 1, h >= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j] -= A[j +
h];
}
vector<ll> Andconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
//XOR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) {
                ll a1 = a[j + h], a0 = a[j];
                a[j + h] = a0 - a1, a[j] = a0 + a1;
            }
}
void ifwht(vector<ll> &a) {
    fwht(A);
    int n = A.size();
    for (int i = 0; i < n; i++) A[i] /= n;
}
vector<ll> Xorconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
```

4.6 gauss

```
const int N = 3e5 + 9;
const double eps = 1e-9;
int Gauss(vector<vector<double>> a, vector<double> &ans) {
    int n = (int)a.size(), m = (int)a[0].size() - 1;
    vector<int> pos(m, -1);
    double det = 1;
```

```
int rank = 0;
for (int col = 0, row = 0; col < m && row < n; ++col) {
    int mx = row;
    for (int i = row; i < n; i++)
        if (fabs(a[i][col]) > fabs(a[mx][col])) mx = i;
    if (fabs(a[mx][col]) < eps) {
        det = 0;
        continue;
    }
    for (int i = col; i <= m; i++) swap(a[row][i], a[mx]
[i]);
    if (row != mx) det = -det;
    det *= a[row][col];
    pos[col] = row;
    for (int i = 0; i < n; i++) {
        if (i != row && fabs(a[i][col]) > eps) {
            double c = a[i][col] / a[row][col];
            for (int j = col; j <= m; j++) a[i][j] -=
a[row][j] * c;
        }
        ++row;
        ++rank;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++) {
        if (pos[i] != -1) ans[i] = a[pos[i]][m] / a[pos[i]]
[i];
    }
    for (int i = 0; i < n; i++) {
        double sum = 0;
        for (int j = 0; j < m; j++) sum += ans[j] * a[i][j];
        if (fabs(sum - a[i][m]) > eps) return -1; // no
solution
    }
    for (int i = 0; i < m; i++)
        if (pos[i] == -1) return 2; // infinte solutions
    return 1; // unique solution
}
```

4.7 geo2d

```
struct point {
    ll x, y;
    point() {}
    point(ll _x, ll _y) : x(_x), y(_y) {}
    point operator+(const point &p) const {
        return {x + p.x, y + p.y};
    }
    point operator-(const point &p) const {
        return {x - p.x, y - p.y};
    }
    bool operator==(const point &p) const {
        return x == p.x && y == p.y;
    }
}
```

```

    bool operator<(const point &p) const {
        if (x == p.x) return y < p.y;
        return x < p.x;
    }
};
// Producto punto
long long dot(const point &a, const point &b) {
    return a.x * b.x + a.y * b.y;
}
// Producto cruzado (determinante)
long long cross(const point &a, const point &b) {
    return a.x * b.y - a.y * b.x;
}
// Cross entre vectores AB y AC
long long cross(const point &a, const point &b, const point &c) {
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x - a.x);
}
// El punto p estÃ en el segmento ab?
bool onSegment(const point &a, const point &b, const point &p) {
    if (cross(a, b, p) != 0) return false;
    return min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x) &&
        min(a.y, b.y) <= p.y &&
        p.y <= max(a.y, b.y);
}
// Interseccion de 2 segmentos
bool intersect(const point &p1, const point &q1, const point &p2, const point &q2) {
    long long d1 = cross(p1, q1, p2);
    long long d2 = cross(p1, q1, q2);
    long long d3 = cross(p2, q2, p1);
    long long d4 = cross(p2, q2, q1);
    if ((d1 > 0 && d2 < 0 || d1 < 0 && d2 > 0) && (d3 > 0 && d4 < 0 || d3 < 0 && d4 > 0))
        return true;
    if (d1 == 0 && onSegment(p1, q1, p2)) return true;
    if (d2 == 0 && onSegment(p1, q1, q2)) return true;
    if (d3 == 0 && onSegment(p2, q2, p1)) return true;
    if (d4 == 0 && onSegment(p2, q2, q1)) return true;
    return false;
}
bool pointInPolygon(const vector<point> &poly, const point &p) {
    bool inside = false;
    int n = poly.size();
    for (int i = 0, j = n - 1; i < n; j = i++) {
        const point &a = poly[i];
        const point &b = poly[j];
        if (onSegment(a, b, p)) return true;
        if ((a.y > p.y) != (b.y > p.y)) {
            long long x_intersection = (b.x - a.x) * (p.y - a.y) - (b.y - a.y) * (p.x - a.x);
            long long y_diff = b.y - a.y;

```

```

        bool right = (x_intersection * y_diff) < 0;
        if (right) inside = !inside;
    }
    return inside;
}

```

4.8 matrix

```

#define matrix array<array<int, N>, N>
matrix multiply(matrix &a, matrix &b) {
    matrix ans{};
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            for (int k = 0; k < N; k++) ans[i][j] =
                smod(ans[i][j], mult(a[i][k], b[k][j]));
    return ans;
}
matrix pow(matrix &a, ll e) {
    matrix ans{};
    for (int i = 0; i < N; i++) ans[i][i] = 1;
    while (e) {
        if (e & 1) ans = multiply(ans, a);
        a = multiply(a, a);
        e >>= 1;
    }
    return ans;
}

```

4.9 millerrabin polardrho

```

ll mult(ll a, ll b, ll mod) {
    return (int128)a * b % mod;
}
ll qp(ll a, ll e, ll mod) {
    ll ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
bool compositetest(ll a, ll d, ll s, ll n) {
    ll x = qp(a, d, n);
    if (x == 1 || x == n - 1) return 0;
    for (ll i = 0; i < s - 1; i++) {
        x = mult(x, x, n);
        if (x == n - 1) return 0;
    }
    return 1;
}
bool esprimo(ll n) {
    if (n < 2) return 0;
    ll d = n - 1, s = 0;

```

```

    while ((d & 1) == 0) d >>= 1, s++;
    for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) {
        if (n == a) return 1;
        if (compositetest(a, d, s, n)) return 0;
    }
    return 1;
}
mt19937_64
rng(chrono::steady_clock::now().time_since_epoch().count());
ll random(ll a, ll b) {
    return uniform_int_distribution<ll>(a, b)(rng);
}
ll f(ll x, ll c, ll mod) {
    ll m = mult(x, x, mod);
    return m + c >= mod ? m + c - mod : m + c;
}
ll getfactor(ll n) {
    if (n % 2 == 0) return 2;
    if (esprimo(n)) return n;
    while (true) {
        ll y = random(1, n - 1);
        ll c = random(1, n - 1);
        ll m = 64, g = 1, r = 1, q = 1;
        ll x, ys;
        while (g == 1) {
            x = y;
            for (ll i = 0; i < r; i++) y = f(y, c, n);
            ll k = 0;
            while (k < r && g == 1) {
                ys = y;
                ll lim = min(m, r - k);
                for (ll i = 0; i < lim; i++) {
                    y = f(y, c, n);
                    ll diff = x > y ? x - y : y - x;
                    q = mult(q, diff, n);
                }
                g = __gcd(q, n);
                k += m;
            }
            r <<= 1;
        }
        if (g == n) {
            do {
                ys = f(ys, c, n);
                ll diff = x > ys ? x - ys : ys - x;
                g = __gcd(diff, n);
            } while (g == 1);
        }
        if (g > 1 && g < n) return g;
    }
}
vector<pll> fp(ll n) {
    stack<ll> st;

```



```

st.push(n);
vector<pll> ans;
map<ll, ll> cont;
while (!st.empty()) {
    ll x = st.top();
    st.pop();
    if (x == 1) continue;
    if (esprimo(x)) cont[x]++;
    else {
        ll f = getfactor(x);
        st.push(f), st.push(x / f);
    }
}
for (auto [x, f] : cont) ans.pb({x, f});
return ans;
}

```

4.10 ntt

```

const ll MOD = 998244353;
vector<vector<int>> w;
int pow2, root;
int smod(int a, int b, int mod) {return (a + b >= mod ? a + b - mod : a + b);}
int rmod(int a, int b, int mod) {return (a - b < 0 ? a - b + mod : a - b);}
int mult(int a, int b, int mod) {return (ll)a * b % mod;}
int qp(int a, int e, int mod) {
    int ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
void calc(int log, int mod) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <<= 1) {
        w[l].resize(len / 2);
        int wn = qp(root, pow2 / len, mod);
        for (int j = 0; 2 * j < len; j++) {
            if (j & 1) w[l][j] = mult(w[l - 1][j / 2], wn, mod);
            else w[l][j] = w[l - 1][j / 2];
        }
    }
}
void findroot(int mod) {
    pow2 = 1;
    int k = mod - 1;
    while (!(k & 1)) pow2 <<= 1, k >>= 1;
    root = 1;
    while (qp(root, pow2, mod) != 1 || qp(root, pow2 / 2,

```

```

mod) == 1) root++;
}
void ntt(vector<int> &a, bool invert, int mod) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <<= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                int u = a[i + j], v = mult(a[i + j + len / 2], w[l][j], mod);
                a[i + j] = smod(u, v, mod);
                a[i + j + len / 2] = rmod(u, v, mod);
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        int invn = qp(n, mod - 2, mod);
        for (int i = 0; i < n; i++) a[i] = mult(a[i], invn, mod);
    }
}
vector<int> convolution(vector<int> a, vector<int> b, int mod = MOD) {
    int n = 1, log = 0, rn = a.size() + b.size() - 1;
    while (n < a.size() + b.size() - 1) n <<= 1, log++;
    findroot(mod);
    calc(log, mod);
    a.resize(n), b.resize(n);
    ntt(a, 0, mod), ntt(b, 0, mod);
    for (int i = 0; i < n; i++) a[i] = mult(a[i], b[i], mod);
    ntt(a, 1, mod);
    a.resize(rn);
    return a;
}
// 2D-FFT
void ntt2D(vector<vector<int>> &a, bool invert, int mod) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) ntt(a[i], invert, mod);
    vector<int> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        ntt(b, invert, mod);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}
vector<vector<int>> convolution2D(vector<vector<int>> a, vector<vector<int>> b, int mod = MOD) {
    int n = 1, logn = 0, rn = a.size() + b.size() - 1;
    while (n < rn) n <<= 1, logn++;

```

```

    int m = 1, logm = 0, rm = a[0].size() + b[0].size() - 1;
    while (m < rm) m <<= 1, logm++;
    findroot(mod);
    calc(max(logn, logm), mod);
    a.resize(n), b.resize(n);
    for (int i = 0; i < n; i++) a[i].resize(m), b[i].resize(m);
    ntt2D(a, 0, mod), ntt2D(b, 0, mod);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) a[i][j] = mult(a[i][j], b[i][j], mod);
    ntt2D(a, 1, mod);
    a.resize(rn);
    for (int i = 0; i < rn; i++) a[i].resize(rm);
    return a;
}

```

4.11 primitive root

```

int primitive_root(int p) {
    int n = p - 1; // n=phi(p) -> n=p-1 si p es primo
    vector<int> fp;
    for (int i = 2; i * i <= n; i++)
        if (n % i == 0) {
            fp.pb(i);
            while (n % i == 0) n /= i;
        }
    if (n > 1) fp.pb(n);
    n = p - 1;
    bool ok = 0;
    for (int g = 2; g <= n; g++) {
        bool ok = 1;
        for (int x : fp)
            if (qp(g, n / x, p) == 1) {
                ok = 0;
                break;
            }
        if (ok) return g;
    }
    return 0;
}

```

4.12 random

```

static mt19937 rng(chrono::steady_clock::now()
    .time_since_epoch().count());
#define rnd(a,b) (uniform_int_distribution<ll>(a,b)(rng))

```

4.13 theorems

4.13.1 Combinatoria

4.13.1.1 General

- $\sum_{k=0}^n \binom{n-k}{k} = \text{Fib}_{n+1}$

- $\binom{n}{k} = \binom{n}{n-k}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $k\binom{n}{k} = n\binom{n-1}{k-1}$
- $\sum_{i=0}^n \binom{n}{i} = 2^n$

Identidad de Vandermonde:

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Identidad del Palo de Hockey (Hockey-Stick):

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

4.13.1.2 Triángulo de Pascal

- En una fila p donde p es un número primo, todos los términos en esa fila excepto los unos son múltiplos de p .
- **Teorema de Kummer:** Para enteros $n \geq m \geq 0$ y un número primo p , la mayor potencia de p que divide a $\binom{n}{m}$ es igual al número de acarreo cuando m se suma a $n - m$ en base p .

4.13.1.3 Números de Catalan

- Fórmula general: $C_n = \frac{1}{n+1} \binom{2n}{n}$
- Recurrencia: $C_0 = 1, C_1 = 1$ y $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$

Representa, entre otras cosas, el número de secuencias de paréntesis balanceadas o triangulaciones de un polígono convexo.

4.13.1.4 Números de Stirling (Primera y Segunda Especie)

- **Primera especie** (número de permutaciones con k ciclos disjuntos):

$$S(n, k) = (n-1) \cdot S(n-1, k) + S(n-1, k-1)$$

- **Segunda especie** (maneras de particionar un conjunto de n objetos en k subconjuntos no vacíos):

$$S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$$

4.13.1.5 Números de Bell

Cuenta el número total de particiones posibles de un conjunto.

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} \cdot B_k$$

4.13.2 Matemáticas

4.13.2.1 Fórmulas Generales

- $a^k - b^k = (a-b) \cdot (a^{k-1}b^0 + a^{k-2}b^1 + \dots + a^0b^{k-1})$
- $|a-b| + |b-c| + |c-a| = 2(\max(a, b, c) - \min(a, b, c))$

Identidad de Lagrange:

$$\left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) - \left(\sum_{k=1}^n a_k b_k \right)^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i b_j - a_j b_i)^2$$

4.13.2.2 Fórmulas de Vieta

Para un polinomio de grado n : $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ con raíces $r_1, r_2, \dots, r_n$:

- $r_1 + r_2 + \dots + r_n = -\frac{a_{n-1}}{a_n}$
- $r_1 r_2 \dots r_n = (-1)^n \frac{a_0}{a_n}$

4.13.2.3 Sucesión de Fibonacci

- $F_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n-k-1}{k}$
- **Fórmula de Binet:** $F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right)$
- $\sum_{i=1}^n F_i = F_{n+2} - 1$

- $\gcd(F_m, F_n) = F_{\gcd(m, n)}$

4.13.2.4 Ternas Pitagóricas

Fórmula de Euclides para generar ternas donde $a^2 + b^2 = c^2$: Para $m > n > 0$:

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2$$

4.13.3 Teoría de Números

4.13.3.1 Función Totiente de Euler (φ)

- $\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$ donde p son factores primos distintos.
- Si a y b son coprimos, entonces $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$
- $\sum_{d|n} \varphi(d) = n$

4.13.3.2 Función e Inversión de Möbius

$\mu(n)$ toma valores en $\{-1, 0, 1\}$:

- 1 si es libre de cuadrados con cantidad par de factores primos.
- -1 si es libre de cuadrados con cantidad impar de factores primos.
- 0 si contiene algún factor primo al cuadrado.

Teorema de inversión de Möbius: Si

$g(n) = \sum_{d|n} f(d)$, entonces:

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right)$$

4.13.3.3 MCD (GCD) y MCM (LCM)

- $\gcd(a, b) \cdot \text{lcm}(a, b) = |a \cdot b|$
- $\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$
- $\sum_{k=1}^n \gcd(k, n) = \sum_{d|n} d \cdot \varphi\left(\frac{n}{d}\right)$

4.13.4 Operaciones de Bits (Bitwise) y Miscelánea

- $a + b = (a \oplus b) + 2(a \text{ AND } b)$
- $a + b = (a | b) + (a \text{ AND } b)$

- El k -ésimo bit está encendido en x si y solo si $x \bmod 2^k \geq 2^{k-1}$.
- $1 \oplus 2 \oplus 3 \oplus \dots \oplus (4k-1) = 0$ para todo $k \geq 0$.

4.13.4.1 Teorema de Erdős-Gallai

Una secuencia descendente de grados $d_1 \geq d_2 \geq \dots \geq d_n$ puede representar un grafo simple finito si la suma total de grados es par y:

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

para cada $1 \leq k \leq n$

4.14 xor basis

```
template <typename T, int LOG>
struct Basis {
    T basis[LOG] = {0};
    int dim = 0;
    bool insert(T x) { // Formar parte del Basis?
        for (int i = LOG - 1; i >= 0; i--)
            if ((x >> i) & 1) {
                if (basis[i] x ^= basis[i];
                else {
                    basis[i] = x, dim++;
                    return true;
                }
            }
        return false;
    }
    T size() {
        return ((T)1 << dim);
    }
    bool find(T x) { // ¿x está en el span{x1,x2,...}?
        for (int i = LOG - 1; i >= 0; i--) x = min(x, x ^ basis[i]);
        return x == 0;
    }
    T max_xor(T x = 0) { // max (x ^ a / a está en el span{x1,x2,...})
        for (int i = LOG - 1; i >= 0; i--) x = max(x, x ^ basis[i]);
        return x;
    }
    T kth(T k) { // The k-th smaller of span{x1,x2,...}
        if (k < 1 || k > ((T)1 << dim)) return -1;
        T x = 0, cont = ((T)1 << dim);
        for (int i = LOG - 1; i >= 0; i--)
            if (basis[i]) {
                if (k > (cont >> 1)) {
```

```
                if (!(x >> i) & 1) x ^= basis[i];
                k -= (cont >> 1);
            } else if ((x >> i) & 1) x ^= basis[i];
            cont >>= 1;
        }
        return x;
    }
}
```

5 Strings

5.1 aho corasick

```
struct Aho_Corasick{
    int t[N][K], cur;
    int pa[N], go[N][K], lnk[N];
    queue<int> qu;

    int insert(string &s){
        int u = 0;
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0)
                t[u][k] = ++cur, pa[cur] = u;
            u = t[u][k];
        }
        return u;
    }

    void bfs(){
        fill(lnk+1, lnk+cur+1, -1);
        for (qu.push(0); !qu.empty(); ){
            int u = qu.front(); qu.pop();
            for (int k=0; k<K; k++){
                int v = t[u][k];
                if (v == 0) go[u][k] = go[lnk[u]][k];
                else{
                    lnk[v] = u == 0 ? 0 : go[lnk[u]][k];
                    go[u][k] = t[u][k];
                    qu.push(v);
                }
            }
        }
    }
} aho; // by NekoRolly
```

5.2 hashing

```
struct Hashing{
    const int mod = 1e9+9;
    const int p = 367, q = 467;
    int ep[N], eq[N], prp[N], prq[N];

    int P(int a, int b){ return 1ll*a*b%mod;}
```

```
int S(int a, int b){ return (a+b)%mod;}

void build(string &s){
    int n = s.size();
    ep[0] = eq[0] = 1;
    prp[0] = prq[0] = 0;
    for (int i=1; i<=n; i++){
        prp[i] = S(P(prp[i-1], p), s[i-1]);
        prq[i] = S(P(prq[i-1], q), s[i-1]);
        ep[i] = P(ep[i-1], p);
        eq[i] = P(eq[i-1], q);
    }

    ll substr(int i, int len){ i++;
        ll hp = S(prp[i+len-1], mod-P(prp[i-1], ep[len]));
        ll hq = S(prq[i+len-1], mod-P(prq[i-1], eq[len]));
        return hp*mod + hq;
    }
} H1, H2; // by NekoRolly
```

5.3 manacher

```
// d1[i] odd number of palindromes centered at i
// d2[i] even number of palindromes centered at (i, i+1)
void manacher(string &s, vector<int> &d1, vector<int> &d2) {
    int n = s.size();
    d1 = vector<int>(n);
    d2 = vector<int>(n, 0);
    int l = -1, r = -1;
    for (int i = 0; i < n; i++) {
        d1[i] = (i > r) ? 1 : min(d1[l+r-i], r-i+1);
        while (i-d1[i] >= 0 && i+d1[i] < n && s[i-d1[i]] == s[i+d1[i]]) {
            d1[i]++;
        }
        if (i+d1[i]-1 > r) r = i+d1[i]-1, l = i-d1[i]-1;
        l = -1, r = -1;
        for (int i = 0; i < n-1; i++) {
            d2[i] = (i >= r) ? 0 : min(d2[l+r-i-1], r-i);
            while (i-d2[i] >= 0 && i+d2[i]+1 < n && s[i-d2[i]] == s[i+d2[i]+1]) {
                d2[i]++;
            }
            if (i+d2[i] > r && d2[i] > 0) r = i+d2[i], l = i-d2[i]+1;
        }
    }
}
```

5.4 prefix function

```
void prefix_function(string &s,int pf[]){
    int n = s.size(); pf[0] = 0;
    for (int i=1; i<n; i++){
        int j = pf[i-1];
        while (j>0 && s[j] != s[i]) j = pf[j-1];
        pf[i] = j + (s[i] == s[j]);
    }
} // by NekoRolly
```

5.5 suffix array

```
enum Comparison { LESS, EQUAL, GREATER };

Comparison getComparison(int a, int b) {
    if (a < b) return Comparison::LESS;
    if (a > b) return Comparison::GREATER;
    return Comparison::EQUAL;
}

const int ALPH = 256;
struct SuffixArray {
    vector<int> suffixArray;
    vector<int> lcp;
    // lcp[i] = largest common prefix of sa[i] and sa[i + 1] (in sorted list)
    vector<int> pos;

    vector<int> sortCyclic(string &s) { // O(n log n)
        int n = s.size();
        vector<int> p(n), c(n), cnt(max(ALPH, n), 0);
        // p[] : sorted array with the starting index of
        substrings of length 2^k
        // c[] : equivalence class
        // counting sort
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < ALPH; i++) cnt[i] += cnt[i - 1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;

        c[p[0]] = 0;
        int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        // radix sort
        vector<int> pNew(n), cNew(n);
        for (int k = 0; (1 << k) < n; k++) {
            // sort pNew by the second substring
            for (int i = 0; i < n; i++) {
                pNew[i] = p[i] - (1 << k);
                if (pNew[i] < 0) pNew[i] += n;
                cnt[i] = 0;
            }
            // counting sort in the first substring
            for (int i = 0; i < n; i++) cnt[c[pNew[i]]]++;
```

```
        for (int i = 1; i < classes; i++) cnt[i] +=
cnt[i - 1];
        for (int i = n - 1; i >= 0; i--) {
            p[--cnt[c[pNew[i]]]] = pNew[i];
        }
        cNew[p[0]] = 0;
        classes = 1;
        for (int i = 1; i < n; i++) {
            int x = p[i] + (1 << k);
            if (x >= n) x -= n;
            int y = p[i - 1] + (1 << k);
            if (y >= n) y -= n;
            pair<int, int> cur = {c[p[i]], c[x]};
            pair<int, int> prev = {c[p[i - 1]], c[y]};
            if (cur != prev) classes++;
            cNew[p[i]] = classes - 1;
        }
        c = cNew;
    }
    return p;
}

void buildLCP(string &s) { // O(n)
    int n = s.size();
    lcp = vector<int>(n - 1, 0);
    pos = vector<int>(n);
    for (int i = 0; i < n; i++) pos[suffixArray[i]] = i;
    int k = 0;
    for (int r = 0; r < n; r++) {
        if (pos[r] == 0) continue;
        int l = suffixArray[pos[r] - 1];
        while (s[l + k] == s[r + k]) k++;
        lcp[pos[r] - 1] = k;
        k = max(0, k - 1);
    }
}
```

// Suffix array: contain starting indexes of the all the suffixes (sorted)

```
SuffixArray(string &s) { // O(n log n)
    char minChar = 'a' - 1;
    s += minChar;
    suffixArray = sortCyclic(s);
    suffixArray.erase(suffixArray.begin());
    s.pop_back();
    buildLCP(s);
}
```

// return the lcp of the suffixes s[l...n - 1] and s[r...n - 1]

```
/*int getLCP(int l, int r) {
    assert(l != r);
    int minPos = min(pos[l], pos[r]);
    int maxPos = max(pos[l], pos[r]);
    return min(lcp[minPos], ..., lcp[maxPos - 1])
```

```
*/

/*Comparison compare(int l1, int r1, int l2, int r2) {
    // compare two substrings [l1, r1], [l2, r2].
    int curLCP = (l1 == l2) ? suffixArray.size() :
getLCP(l1, l2);
    if (r1 < l1 + curLCP) {
        if (r2 < l2 + curLCP) {
            int sz1 = r1 - l1 + 1;
            int sz2 = r2 - l2 + 1;
            return getComparison(sz1, sz2);
        } else return Comparison::LESS;
    } else if (r2 < l2 + curLCP) return
Comparison::GREATER;
    return getComparison(pos[l1], pos[l2]);
}*/
};
```

5.6 z function

```
void z_function(string &s,int zf[]){
    int n = s.size();
    for (int i=1,l=0,r=0; i<n; i++){
        if (i < r) zf[i] = min(zf[i-l], r-i);
        while (i+zf[i] < n && s[i+zf[i]] == s[zf[i]]+
+;
        if (i+zf[i] > r) l = i, r = i+zf[i];
    }
} // by NekoRolly
```